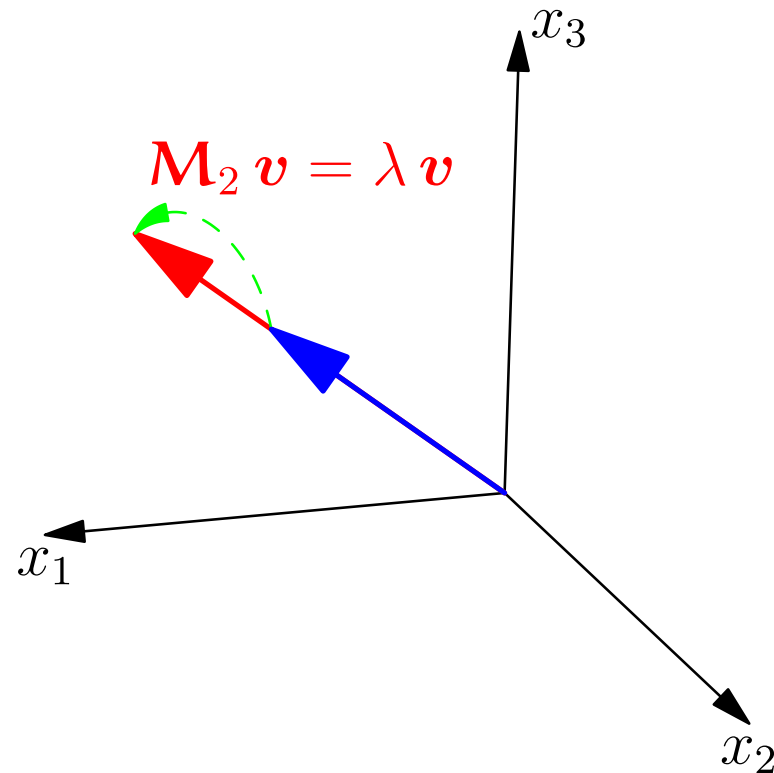


Mathematics of Machine Learning

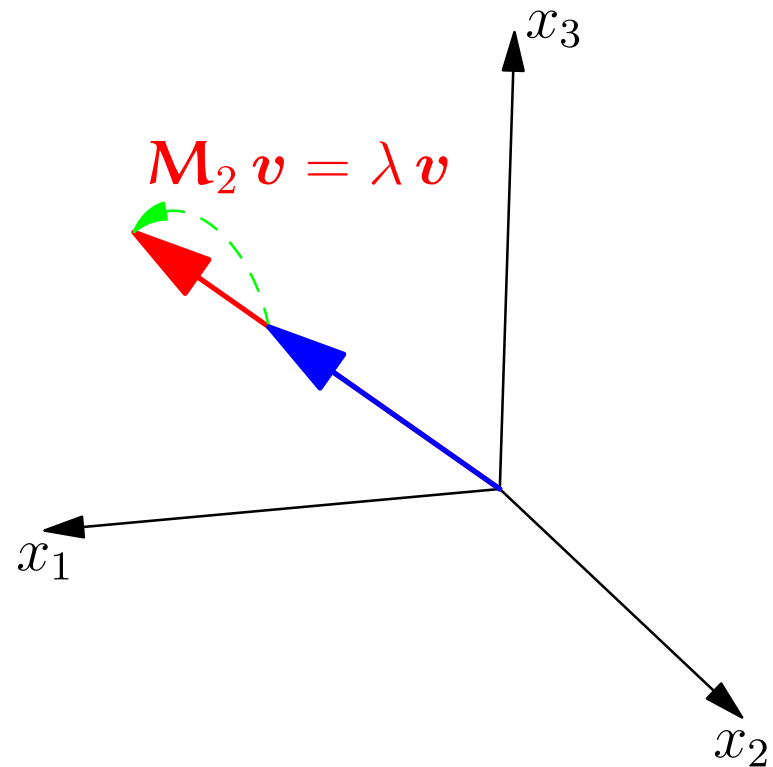
Eigensystems



Eigenvectors, Orthogonal Matrices, Eigenvector Decomposition, Rank

Outline

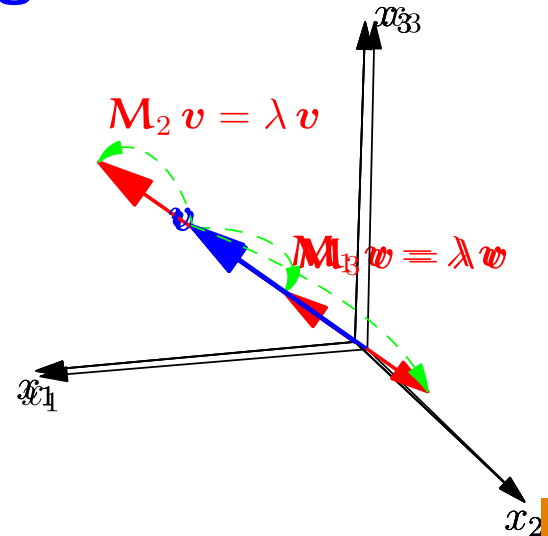
1. **Eigenvectors**
2. Orthogonal Matrices
3. Eigen Decomposition
4. Low Rank Approximation
5. Non-Normal Matrices



Eigenvector equation

- Eigen-systems help us to understand mappings
- A non-zero vector v is said to be an **eigenvector** if

$$Mv = \lambda v$$



- M is square (i.e. $n \times n$)
- Where the number λ is the **eigenvalue**
- Eigenvalues play a fundamental role in understanding operators

Symmetric Matrices

- If $\mathbf{M}$ is an $n \times n$ **symmetric** matrix then it has n real orthogonal eigenvectors with real eigenvalues■
- We denote the i^{th} eigenvector by $\mathbf{v}_i$ and the corresponding eigenvalue by λ_i so that

$$\mathbf{M}\mathbf{v}_i = \lambda_i \mathbf{v}_i \blacksquare$$

- Orthogonal means that if $i \neq j$ then

$$\mathbf{v}_i^T \mathbf{v}_j = 0 \blacksquare$$

- (We can always normalise eigenvectors if we want)■

Proof of Orthogonality

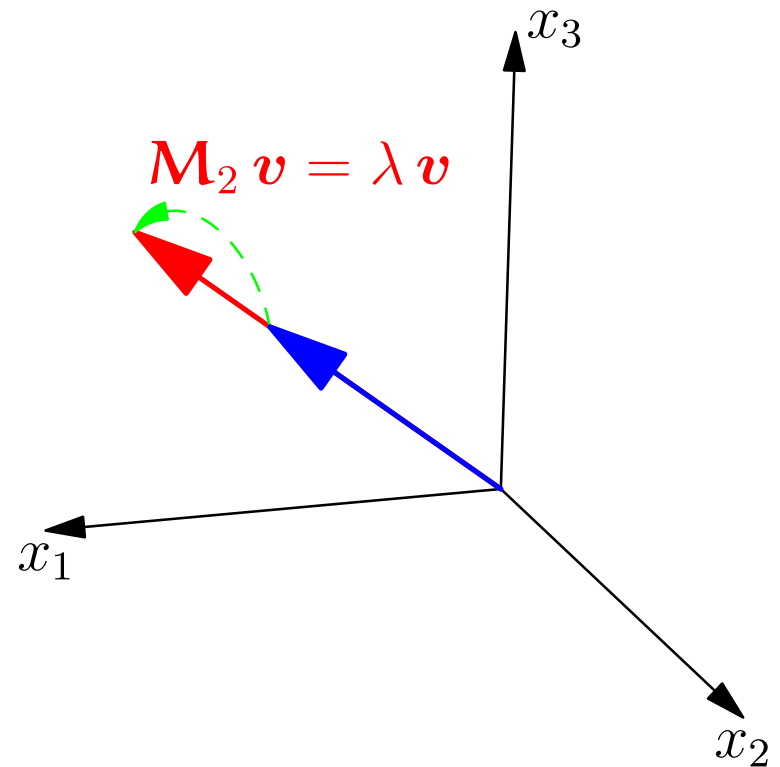
- $(\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i)^\top$ implies $\mathbf{v}_i^\top \mathbf{M}^\top = \lambda_i\mathbf{v}_i^\top$ ■
- When $\mathbf{M}$ is symmetric then $\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i \Rightarrow \mathbf{v}_i^\top \mathbf{M} = \lambda_i\mathbf{v}_i^\top$ ■
- Consider two eigenvectors $\mathbf{v}_i$ and $\mathbf{v}_j$ of $\mathbf{M}$

$$\begin{aligned}\mathbf{v}_i^\top \mathbf{M}\mathbf{v}_j &= (\mathbf{v}_i^\top \mathbf{M})\mathbf{v}_j = \lambda_i\mathbf{v}_i^\top \mathbf{v}_j \\ &= \mathbf{v}_i^\top (\mathbf{M}\mathbf{v}_j) = \lambda_j\mathbf{v}_i^\top \mathbf{v}_j \blacksquare\end{aligned}$$

- So either $\lambda_i = \lambda_j$ or $\mathbf{v}_i^\top \mathbf{v}_j = 0$ ■
- If $\lambda_i = \lambda_j$ then any linear combination of $\mathbf{v}_i$ and $\mathbf{v}_j$ is an eigenvector $(\mathbf{M}(a\mathbf{v}_i + b\mathbf{v}_j) = \lambda_i(a\mathbf{v}_i + b\mathbf{v}_j))$ ■ So I can choose two eigenvectors that are orthogonal to each other. ■

Outline

1. Eigenvectors
2. **Orthogonal Matrices**
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Orthogonal Matrices

- We can construct an **orthogonal** matrix $\mathbf{V}$ from the normalised eigenvectors ($\mathbf{v}_i^\top \mathbf{v}_i = 1$)

$$\mathbf{V} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$$

- Matrix $\mathbf{V}$ is an $n \times n$ matrix
- Because the vectors $\mathbf{v}_i$ are orthogonal and normalised

$$\mathbf{V}^\top \mathbf{V} = \begin{pmatrix} \mathbf{v}_1^\top \mathbf{v}_1 & \mathbf{v}_1^\top \mathbf{v}_2 & \cdots & \mathbf{v}_1^\top \mathbf{v}_n \\ \mathbf{v}_2^\top \mathbf{v}_1 & \mathbf{v}_2^\top \mathbf{v}_2 & \cdots & \mathbf{v}_2^\top \mathbf{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_n^\top \mathbf{v}_1 & \mathbf{v}_n^\top \mathbf{v}_2 & \cdots & \mathbf{v}_n^\top \mathbf{v}_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = \mathbf{I}$$

The Other Way Around

- We have shown that $V^T V = I$
- Thus multiply both sides on the left by V

$$VV^T V = V$$

- V will have an inverse (we show this on the next slide), V^{-1} , such that $VV^{-1} = I$
- Multiplying the equation on the right by V^{-1}

$$(VV^T)VV^{-1} = VV^{-1}$$

$$VV^T = I$$

- Note that, $V^{-1} = V^T$ (definition of orthogonal matrix)

Invertible Matrices

- A matrix, $\mathbf{M}$, will be singular (not invertible) if there exists a vector $\mathbf{x}$ ($\neq \mathbf{0}$) such that

$$\mathbf{M}\mathbf{x} = \mathbf{0}$$

- Now if there exists such a vector such that $\mathbf{V}\mathbf{x} = \mathbf{0}$ then multiplying by $\mathbf{V}^T$ we get

$$\mathbf{V}^T\mathbf{V}\mathbf{x} = \mathbf{V}^T\mathbf{0}$$

$$\mathbf{x} = \mathbf{0}$$

since $\mathbf{V}^T\mathbf{V} = \mathbf{I}$

- Thus $\mathbf{V}$ is invertible

Rotations

- Orthogonal matrices satisfy $\mathbf{V}^\top \mathbf{V} = \mathbf{V} \mathbf{V}^\top = \mathbf{I}$
- As a consequence they define rotations (and possibly a reflection)
- Consider a vector $\mathbf{x}$ and $\mathbf{x}' = \mathbf{V}\mathbf{x}$, now

$$\|\mathbf{x}'\|_2^2 = \mathbf{x}'^\top \mathbf{x}' = (\mathbf{V}\mathbf{x})^\top (\mathbf{V}\mathbf{x}) = \mathbf{x}^\top \mathbf{V}^\top \mathbf{V} \mathbf{x} = \mathbf{x}^\top \mathbf{x} = \|\mathbf{x}\|_2^2$$

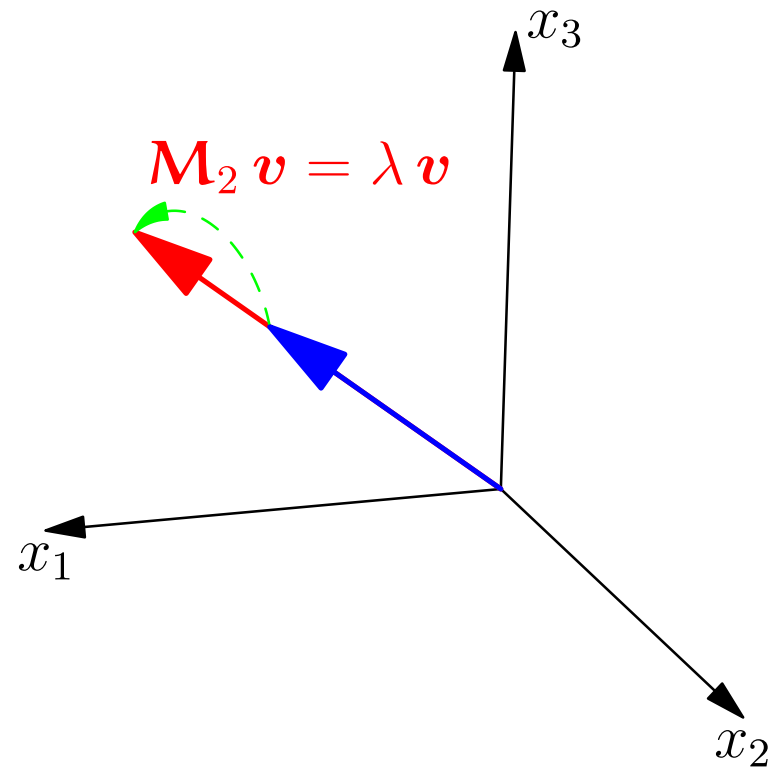
- Similarly if additionally $\mathbf{y}' = \mathbf{V}\mathbf{y}$ then

$$\langle \mathbf{x}', \mathbf{y}' \rangle = (\mathbf{V}\mathbf{x})^\top (\mathbf{V}\mathbf{y}) = \mathbf{x}^\top \mathbf{V}^\top \mathbf{V} \mathbf{y} = \mathbf{x}^\top \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos(\theta)$$

- Rotations and reflections preserve lengths and angles

Outline

1. Eigenvectors
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Matrix Decomposition

- Taking the matrix of eigenvectors, V , then

$$MV = M(v_1, v_2, \dots, v_n) = (\lambda_1 v_1, \lambda_2 v_2, \dots, \lambda_n v_n) = V\Lambda$$

- where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$

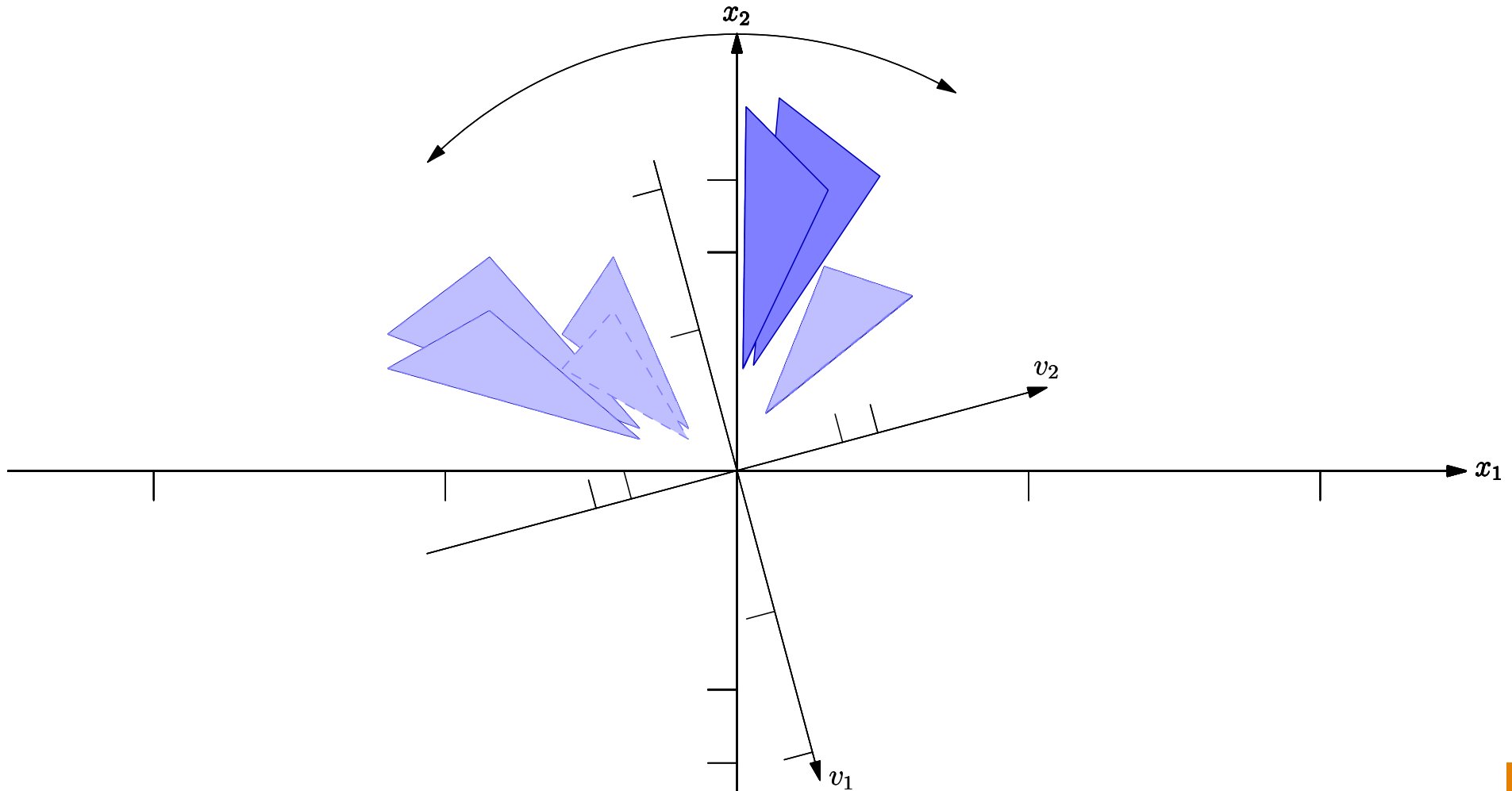
- Now

$$M = MVV^T = V\Lambda V^T$$

- Very important *similarity transform*

Mappings by Symmetric Matrices

$$\mathbf{M} = \begin{pmatrix} 0.83 & -0.31 \\ -0.31 & 1.9 \end{pmatrix} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T = \begin{pmatrix} \cos(-75^\circ) & -\sin(-75^\circ) \\ \sin(-75^\circ) & \cos(-75^\circ) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75^\circ) & -\sin(75^\circ) \\ \sin(75^\circ) & \cos(75^\circ) \end{pmatrix}$$



Inverses

- For any symmetric invertible matrix

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$$

$$\mathbf{M}^{-1} = \mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T$$

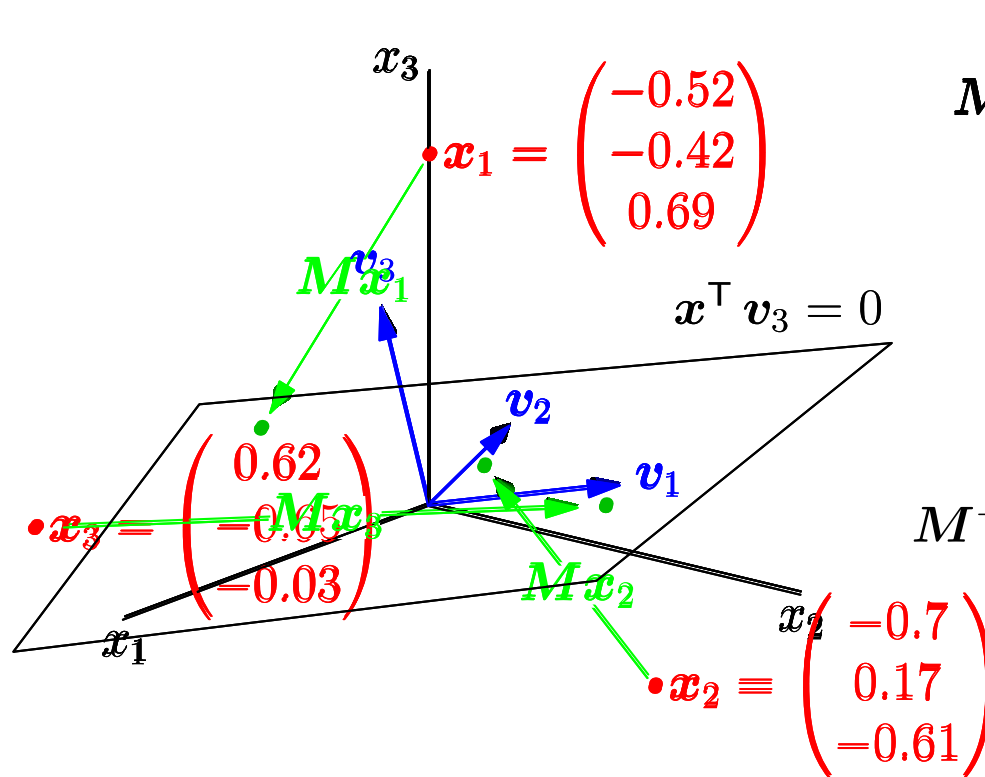
- Where $\mathbf{\Lambda}^{-1} = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \dots & 0 \\ 0 & \frac{1}{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{\lambda_n} \end{pmatrix}$

- Since

$$\begin{aligned} \mathbf{M}\mathbf{M}^{-1} &= (\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T)(\mathbf{V}\mathbf{\Lambda}^{-1}\mathbf{V}^T) = \mathbf{V}\mathbf{\Lambda}(\mathbf{V}^T\mathbf{V})\mathbf{\Lambda}^{-1}\mathbf{V}^T \\ &= \mathbf{V}\mathbf{\Lambda}\mathbf{\Lambda}^{-1}\mathbf{V}^T = \mathbf{V}\mathbf{V}^T = \mathbf{I} \end{aligned}$$

- I.e. small eigenvalues become large eigenvalues and vice versa

III-Conditioning Again



$$M = \begin{pmatrix} 0.80 & 0.01 & 0.25 \\ 0.01 & 0.85 & 0.48 \\ 0.25 & 0.48 & 0.5 \end{pmatrix}$$

$$= V \begin{pmatrix} 0.8 & 0 & 0 \\ 0 & 1.25 & 0 \\ 0 & 0 & 0.1 \end{pmatrix} V^T$$

$$M^{-1} = \begin{pmatrix} 1.9 & 1.17 & -2.1 \\ 1.17 & 3.34 & -3.8 \\ -2.1 & -3.8 & 6.8 \end{pmatrix}$$

$$= V \begin{pmatrix} 1.25 & 0 & 0 \\ 0 & 0.8 & 0 \\ 0 & 0 & 10 \end{pmatrix} V^T$$

Condition Number

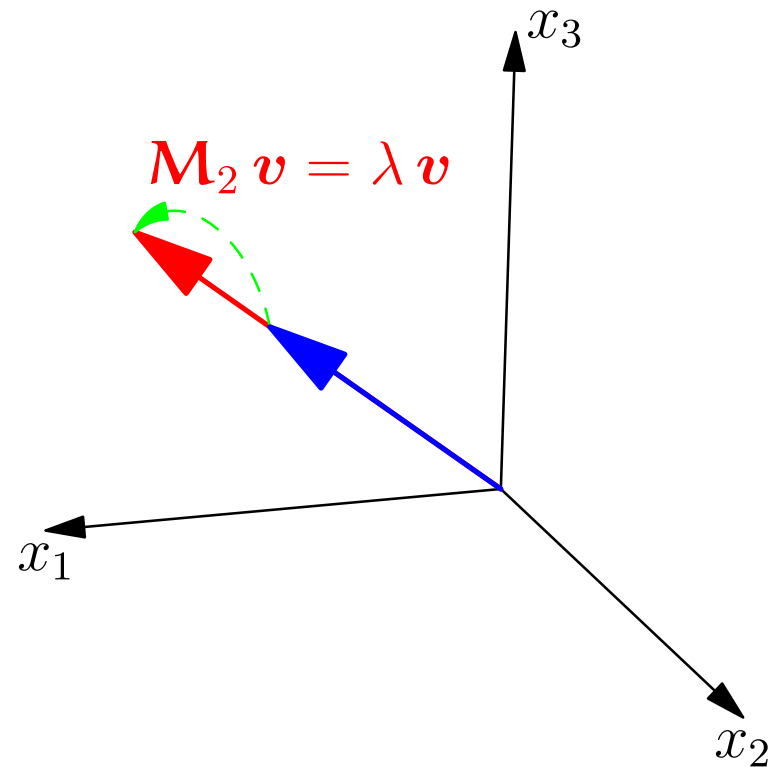
- Taking matrix inverses can be inherently unstable■
- Any small error can be amplified by taking the inverse■
- The stability of the inverse depends on the ratio of smallest eigenvalue to the largest eigenvalue (i.e. the biggest possible amplification compared to the smallest)■
- For symmetric matrices the largest $|\lambda|$ is the spectral norm, $\|\mathbf{M}\|_2$, from the vector spaces lecture■
- The condition number is given by

$$\|\mathbf{M}\|_2 \times \|\mathbf{M}^{-1}\|_2 = \frac{|\lambda_{\max}|}{|\lambda_{\min}|} \blacksquare$$

- Large condition number implies very ill-conditioned■

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Rank of a Matrix

- The rank of a symmetric matrix, $\mathbf{M}$, is the number of non-zero eigenvalues■
- The space spanned by the eigenvectors $\mathbf{v}_a$, $\mathbf{v}_b$, etc. with zero eigenvalue forms a **null space**■
- Any vector in the null space will get mapped to the zero vector

$$\mathbf{M}(a\mathbf{v}_a + b\mathbf{v}_b + \dots) = \mathbf{0}■$$

- A square matrix is said to be **rank deficient** if it has any eigenvectors with eigenvalue equal to 0■
- This happens when the columns of the matrix are not linearly independent■

“Inverting” Rank Deficient Matrices

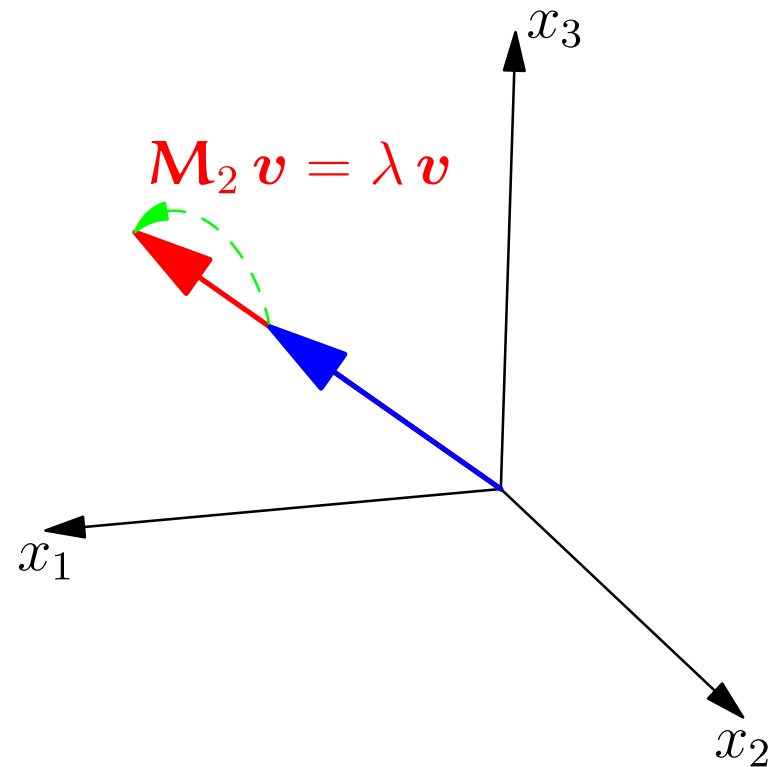
- Rank deficient matrices are non-invertible (i.e. we don't know the vector x such that $\mathbf{M}x = b$) as we don't know the component of the x in the null space■
- Although we don't know x we can find a vector, x , that satisfies $\mathbf{M}x = b$ (provided one exists)■
- Given a symmetric $n \times n$ matrix with k non-zero eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ we can construct a “pseudo-inverse” $\mathbf{M}^+$ as $\mathbf{V}\mathbf{\Lambda}^+\mathbf{V}^T$ where $\mathbf{\Lambda}^+ = \text{diag}(\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_k}, 0, \dots, 0)$ ■
- This finds the vector x with no component in the null space■ (it is the solution with the smallest norm)■
- This and $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ are special cases of the pseudo-inverse we meet with SVD■

Low Rank Approximation

- Recall that matrices with large and small eigenvalues are ill-conditioned so the inverse has the potential to greatly amplify any measurement error■
- One work around is to set all small eigenvalues to zero and use the pseudo-inverse■
- Setting small eigenvalues to zero reduces the rank of the matrix and is an example of a low rank approximation■
- Low rank approximations are much used to obtain approximate models for arrays of data■ (we will revisit this when we look at SVD)■

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Non-Normal Matrices

- Symmetric matrices are part of a class of matrices known as **normal matrices**, which satisfy $\mathbf{M}\mathbf{M}^\dagger = \mathbf{M}^\dagger\mathbf{M}$ (for real matrices $\mathbf{M}^\dagger = \mathbf{M}^\top$)
- Hermitian matrices, $\mathbf{H}$, are complex matrices that equal their conjugate transpose: $\mathbf{H} = (\mathbf{H}^\top)^* = \mathbf{H}^\dagger$
- Hermitian matrices are also normal; they are important in quantum theory, less so in machine learning
- Normal matrices are exactly those with a full set of orthonormal (possibly complex) eigenvectors
- Most non-symmetric real matrices are not normal (rotations are an exception); their eigenvectors are more confusing

Asymmetric Matrices

- For $n \times n$ real symmetric matrices we are guaranteed there are n orthogonal real eigenvectors■
- For non-symmetric matrices the eigenvectors will generally not be orthogonal, they won't necessarily be real■
- We are not even guaranteed there will be n linearly independent eigenvectors■
- Matrices which don't have n linearly independent eigenvectors are said to be **defective**, e.g. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only one■
- Randomly generated matrices will typically not be defective, but nevertheless they can arise in certain situations■

Characteristic Equation

- Since our eigenvectors are defined by $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ then

$(\mathbf{M} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ has a solution $\mathbf{v} \neq \mathbf{0}$ only if $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$ ■

- This is known as the characteristic equation and leads to an n^{th} order polynomial in λ ■
- An n^{th} order polynomial has n roots (counting repeats), which are in general complex■
- Thus even for real (non-symmetric) matrices we can get complex eigenvalues■

Complex Conjugate Pairs

- If M is real and λ is complex then v must be complex too, otherwise Mv would be real but λv would not
- Furthermore

$$Mv = \lambda v \quad \implies \quad Mv^* = \lambda^* v^*$$

- So that eigenvectors and their corresponding eigenvalues exist in complex conjugate pairs

Left and Right Eigenvectors

- For symmetric matrices if $\mathbf{v}$ is a right-hand eigenvector in the sense that $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$ then $\mathbf{v}^\top$ is a left-hand eigenvector

$$\mathbf{v}^\top \mathbf{M} = \lambda \mathbf{v}^\top$$

- Earlier we used this property to show that the eigenvectors for symmetric matrices are orthogonal
- For asymmetric real matrices the left-hand eigenvectors $\mathbf{u}^\top$ are different from the right-hand eigenvectors, although they share the same eigenvalues (they have the same characteristic equation)

$$\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$$

$$\mathbf{u}^\top \mathbf{M} = \lambda \mathbf{u}^\top$$

- As before, $\mathbf{u}_i^\top \mathbf{v}_j = 0$ if $\lambda_i \neq \lambda_j$, but in general $\mathbf{v}_i^\top \mathbf{v}_j \neq 0$

Eigen-Decomposition

- For a non-symmetric matrix, $\mathbf{M}$, provided it is not defective, we can decompose the matrix as

$$\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$$

- $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues
- $\mathbf{V}$ is a matrix whose columns are the right-hand eigenvectors—in general it is not an orthogonal matrix
- $\mathbf{V}^{-1}$ is a matrix whose rows are the left-hand eigenvectors (scaled so that $\mathbf{u}_i^T \mathbf{v}_i = 1$)

Summary

- Linear mappings are commonly used in machine learning algorithms such as regression■
- We can understand symmetric operators by looking at their eigenvectors■
- Any symmetric matrix can be decomposed as $\mathbf{M} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$
 - ★ where $\mathbf{V}$ is an orthogonal matrix whose columns are the eigenvectors
 - ★ and $\mathbf{\Lambda}$ is a diagonal matrix of the eigenvalues■
- This decomposition allows us to understand inverse mappings■
- Non-symmetric matrices are more complex■—we use the eigen-decomposition less because of this■